So, you write a small piece of code that automatically performs the steps above for each order, and checks all the distances for each order.

After another year, however, the business is going so well that your CEO decides you are ready to go national, after closing a deal that will see hundreds or thousands of medium and large stores (spread across the Country) join your platform.

Computing millions of distances per user start seeming a little overwhelming and inefficient – also, since you are only in the late 90s, servers are not yet that fast, server farms are a rarity and data centers are something for large hardware companies like IBM, they are not yet a thing for e-commerce.

8.2.1 First Attempts

Your first proposal can be precomputing the closest warehouse for each user once and for all products, but that doesn't really work, as users can and will move, sometimes they want to have stuff sent to their office or to the post office, not just to their homes. Plus, the availability of goods will change in time, so the closest shop isn't always the best one. You would need to keep a list of shops, ordered by distance, for each customer (or at least each city): haven't I already mentioned that data centers aren't yet a thing?

8.2.2 Sometimes Caching is NOT the Answer